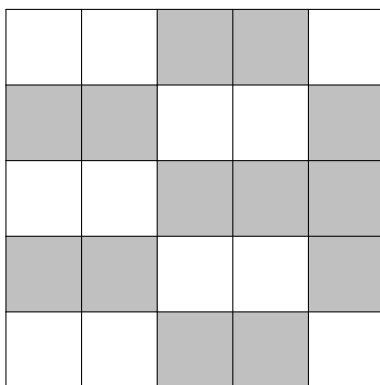


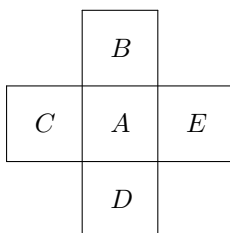
Problem 77 (Belarus TST 2017 Test 6 Problem 2). Any cell of a 5×5 table is coloured black or white. Find the greatest possible value of the ways to place a T -tetramino on the table so that it covers exactly two white and two black cells (for various colourings of the table).

Solution. There are at most 37 ways to place the T -tetramino. The following figure provides an example for which there are 37 ways to place it by routine checking.



Consider 5 cells forming a cross shape as shown below. There are 4 T -tetraminoes having A as the centre. We claim that at most 3 of these T -tetraminoes are good (meaning that it covers exactly two white cells and two black cells). Suppose on the contrary that all 4 T -tetraminoes are good. Since each of $\{A, B, C, D\}$ and $\{A, B, C, E\}$ contains exactly 2 white cells, the colours of D and E must be the same. Similarly, the colours of B, C, D, E are the same. But then $\{A, B, C, D\}$ contains at least 3 cells B, C, D of the same colour, contradiction.

Also, in order that exactly 3 T -tetraminoes are good, exactly one of B, C, D, E has the same colour as A .



Now, each of the 9 cells not on the boundary is the centre of a cross, while each of the 12 cells on the boundary but not a corner is the centre of exactly one T -tetramino. It follows that there are at most

$$3 \times 9 + 12 = 39$$

good T -tetraminoes. Suppose there are at least 38 ways to place the T -tetramino. We want to derive a contradiction. We say that a cell not on the boundary is special if it has the same colour as exactly one of its adjacent cells, and we say that a cell on the boundary but not a corner is special if it is the centre of a good T -tetramino. Note that there is at most one non-corner cell which is not special. WLOG assume the possibly non-special cell lies in the bottom right 3×3 square. In the following, we use (i, j) to mean the cell in row i (counted from the top) and column j (counted from the left). WLOG assume $(1, 1)$ is white.

- Case 1: Both $(1, 2)$ and $(2, 1)$ are white.
 - As $(1, 2)$ and $(2, 1)$ are special, all of $(1, 3), (2, 2), (3, 1)$ are black.
 - As $(2, 3)$ and $(3, 2)$ are special, they must be white.
 - But then $(2, 2)$ is not special, contradiction.
- Case 2: $(1, 2)$ is white but $(2, 1)$ is black (or vice versa)
 - As $(1, 2)$ and $(2, 1)$ are special, $(1, 3)$ and $(2, 2)$ are black, and $(3, 1)$ is white.
 - As $(2, 2)$ is special, $(2, 3)$ and $(3, 2)$ are white.
 - As $(1, 3)$ and $(3, 1)$ are special, $(1, 4)$ and $(4, 1)$ are black.
 - As $(2, 3)$ and $(3, 2)$ are special, $(4, 2)$ and $(3, 3)$ are black, and $(2, 4)$ is white.
 - As $(1, 4)$ and $(4, 1)$ are special, $(1, 5)$ and $(5, 1)$ are white.
 - As $(2, 4)$ and $(4, 2)$ are special, $(2, 5)$ and $(3, 4)$ are black, while $(4, 3)$ and $(5, 2)$ are white.
 - As $(2, 5)$ and $(5, 2)$ are special, $(3, 5)$ and $(5, 3)$ are black.
 - Now, note that both $(3, 4)$ and $(3, 5)$ are not special, contradiction.
- Case 3: Both $(1, 2)$ and $(2, 1)$ are black.
 - As $(2, 2)$ is special, it must be white.
 - As $(1, 2)$ and $(2, 1)$ are special, $(1, 3)$ and $(3, 1)$ are black.
 - As $(1, 3)$ and $(3, 1)$ are special, all of $(1, 4), (2, 3), (3, 2), (4, 1)$ are white.
 - But then $(2, 2)$ is not special, contradiction.

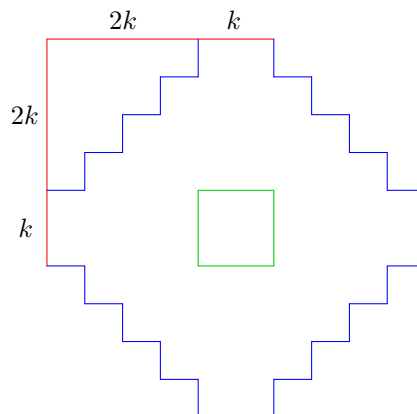
This proves there are at most 37 ways to place the T -tetramino. □

Problem 78 (USA TST for EGMO 2020). Vulcan and Neptune play a turn-based game on an infinite grid of unit squares. Before the game starts, Neptune chooses a finite number of cells to be flooded. Vulcan is building a levee, which is a subset of unit edges of the grid (called walls) forming a connected, non-self-intersecting path or loop (*).

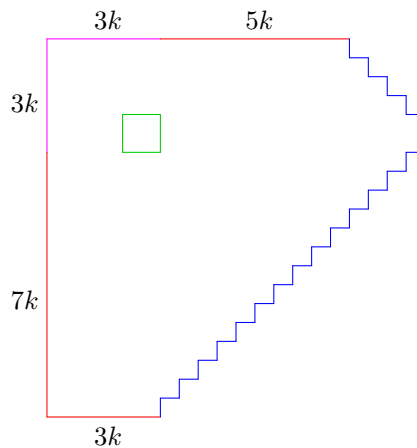
The game then begins with Vulcan moving first. On each of Vulcans turns, he may add up to three new walls to the levee (maintaining the conditions for the levee). On each of Neptunes turns, every cell which is adjacent to an already flooded cell and with no wall between them becomes flooded as well. Prove that Vulcan can always, in a finite number of turns, build the levee into a closed loop such that all flooded cells are contained in the interior of the loop, regardless of which cells Neptune initially floods.

* More formally, there must exist lattice points $A_0, A_1, \dots, A_k$, pairwise distinct except possibly $A_0 = A_k$, such that the set of walls is exactly $\{A_0A_1, A_1A_2, \dots, A_{k-1}A_k\}$. Once a wall is built it cannot be destroyed; in particular, if the levee is a closed loop (i.e. $A_0 = A_k$) then Vulcan cannot add more walls. Since each wall has length 1, the length of the levee is k .

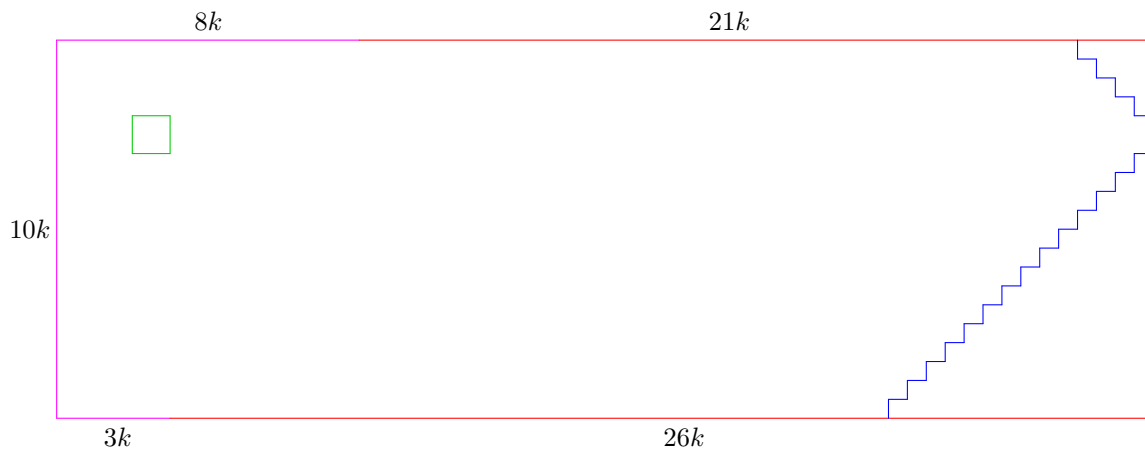
Solution. Suppose the flooded cells lie inside a $k \times k$ green square at the beginning. In the first $2k$ rounds, Vulcan can build the red walls as shown. At this moment, the flooded cells are bounded by the blue region.



Then in each of the next $5k$ rounds, Vulcan keeps on extending the wall at the top by 1 unit, and the wall at the bottom left by 2 units in the way as shown below. The flood is always blocked by the walls at the top and on the left. At the end, the flood just reaches the wall at the bottom since $2k + 5k = 7k$ rounds have passed. Thus, the flooded cells are bounded by the blue region.



Lastly, in each of the next $19k$ rounds, Vulcan keeps on extending the wall at the top and at the bottom by at least 1 unit each (and 3 units in total) in the way as shown below. Then the flood is always blocked by the walls at the top and on the left. At the end, the flood just reaches the wall on the right since $7k + 19k = 26k$ rounds have passed. Thus, the flooded cells are bounded by the blue region, and hence by the walls.



□

Problem 79 (Croatia TST 2016 Problem 2). Let N be a positive integer. Consider an $N \times N$ array of square unit cells. Two corner cells that lie on the same longest diagonal are coloured black, and the rest of the array is white. A move consists of choosing a row or a column and changing the colour of every cell in the chosen row or column.

What is the minimal number of additional cells that one has to colour black such that, after a finite number of moves, a completely black board can be reached?

Solution. Equivalently, we need to minimize the number of white cells after applying some moves to the initial array, given that the two corner cells (say, the top left and the bottom right cells) remain black at the end. Note that the order of applying the moves does not matter. Also, there is no effect if we apply two moves to the same row or column. Therefore, WLOG assume we have applied a move to each of a rows (not counting the first and the last), and to each of b columns (not counting the first and the last).

- If a move is applied to each of the first row and the last row, we must apply a move to each of the first column and the last column in order that the two corner cells are black. Note that this uniquely determines the colour of each cell at the end. Indeed, for each cell, we just need to check whether moves are applied to its row and its column.
 - The top right corner cell and the bottom left corner cell are white.
 - There are $2a$ white cells on the left edge and on the right edge (excluding corner cells).
 - There are $2b$ white cells on the top edge and on the bottom edge.
 - There are $ab + (n - 2 - a)(n - 2 - b)$ white cells in the middle.

Therefore, the number of white cells is

$$N = 2 + 2a + 2b + ab + (n - 2 - a)(n - 2 - b).$$

As it is linear and symmetric in a and b , it suffices to check $(a, b) = (0, 0), (0, n - 2), (n - 2, n - 2)$ to seek for the minimum value. The corresponding values are $N = n^2 - 4n + 6, 2n - 2, n^2 - 2$, all of which are larger than $2n - 4$.

- If a move is applied to the first row but not the last row (or vice versa), we must apply a move to the first column but not the last column. Similar to the first case, the number of white cells is

$$N = (n - 2) + (n - 2) + ab + (n - 2 - a)(n - 2 - b).$$

By checking $(a, b) = (0, 0), (0, n - 2), (n - 2, n - 2)$, we find that the minimum value of N in this case is $2n - 4$, attained when $(a, b) = (0, n - 2)$ or its permutation.

- If no move is applied to the first and the last rows, no move is applied to the first and the last columns. The number of white cells is

$$N = 2 + 2(n - 2 - a) + 2(n - 2 - b) + ab + (n - 2 - a)(n - 2 - b).$$

This is symmetric to the first case under the mapping $(a, b) \mapsto (n - 2 - a, n - 2 - b)$. Therefore, we still have $N > 2n - 4$.

Thus, the minimum value of N is $2n - 4$. □

Problem 80 (British MO 2020 Round 2 Problem 3). A 2019×2019 square grid is made up of 2019^2 unit cells. Each cell is coloured either black or white. A colouring is called *balanced* if, within every square subgrid made up of k^2 cells for $1 \leq k \leq 2019$, the number of black cells differs from the number of white cells by at most one. How many different balanced colourings are there?

(Two colourings are different if there is at least one cell which is black in exactly one of them.)

Solution. There are $2^{2012} - 6$ balanced colourings in total.

A row or a column is said to be good if the cells in that row or column are coloured black and white alternately. We first show that all rows are good or all columns are good. This is obvious if no two adjacent cells have the same colour. If there are two adjacent cells (say, in the same row) which are of the same colour, by considering some 2×2 square, we see the two cells above them are of different colours to them. The same holds for the two cells below them. Inductively, we see the two columns are good. Now, if not every column is good, we can find two adjacent columns such that one is good and one is not. Then there must be two adjacent cells in these columns having the same colour. But then we have seen that this implies both columns are good, contradiction. This proves the claim.

WLOG assume all rows are good. Then the colouring is uniquely determined by the colouring in the first column. For even k , any $k \times k$ square must contain the same number of white and black cells since each row does. For odd k , the numbers of white and black cells in a $k \times k$ square differs by 1 if and only if the numbers of white and black cells in the corresponding cells in the first column differ by 1. So it remains to count the number of ways to colour the first column such that the numbers of white and black cells in any $2m - 1$ consecutive cells differ by 1 (for $m = 1, 2, \dots, 1010$).

We draw a line between any two adjacent cells of the same colour. If there is an odd number of cells $C_{t+1}, C_{t+2}, \dots, C_{t+2m-1}$ between two lines, WLOG assume C_{t+1} is black. Then the number of black cells among $C_t, C_{t+1}, \dots, C_{t+2m}$ is 3 more than the number of white cells, contradiction.

Conversely, if there is an even number of cells between any two lines, consider any odd number of consecutive cells. Since the numbers of white and black cells between any two adjacent lines are the same, we may assume there is at most one line in this group of cells. Then it is clear that the numbers of white and black cells differ by exactly 1. So the condition is satisfied.

If lines are only added between the $(2j - 1)$ st cell and the $(2j)$ th cell for some j , there are 2^{1009} ways to add the lines. If lines are only added between the $(2j)$ th cell and the $(2j + 1)$ st cell for some j , there are 2^{1009} ways to add the lines. But then there is 1 case where no line is added, and this case is counted twice. Since there are 2 ways to colour the cells after the lines are added (depending on the colour of the first cell), there are

$$2(2 \times 2^{1009} - 1) = 2^{1011} - 2$$

colourings of the first column.

Lastly, we can swap the roles of rows and columns. Note that the 2 chessboard colourings are counted twice. Therefore, the number of balanced colourings is

$$2(2^{1011} - 2) - 2 = 2^{1012} - 6.$$

□